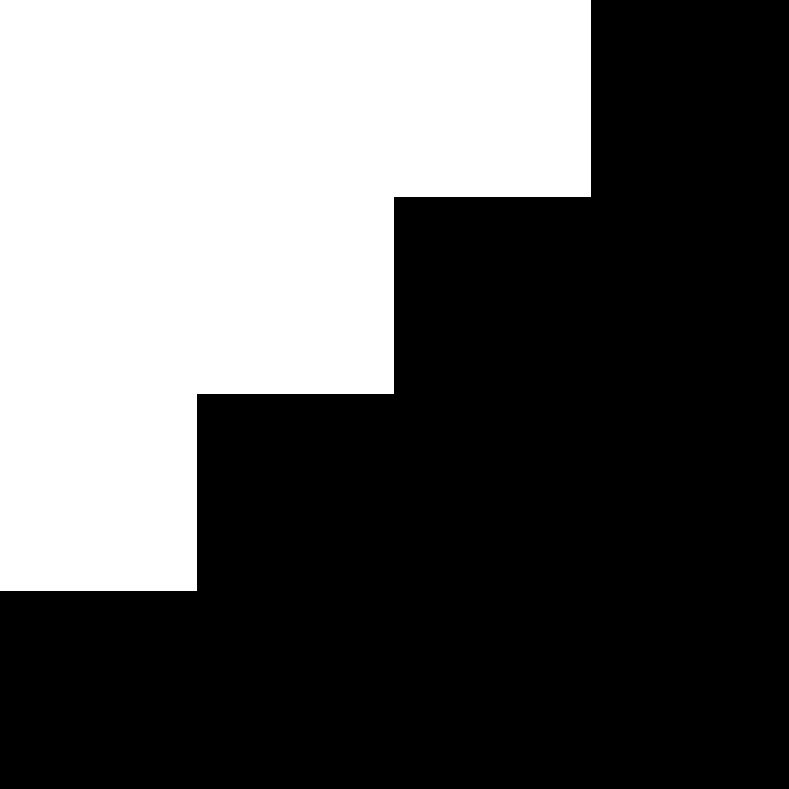
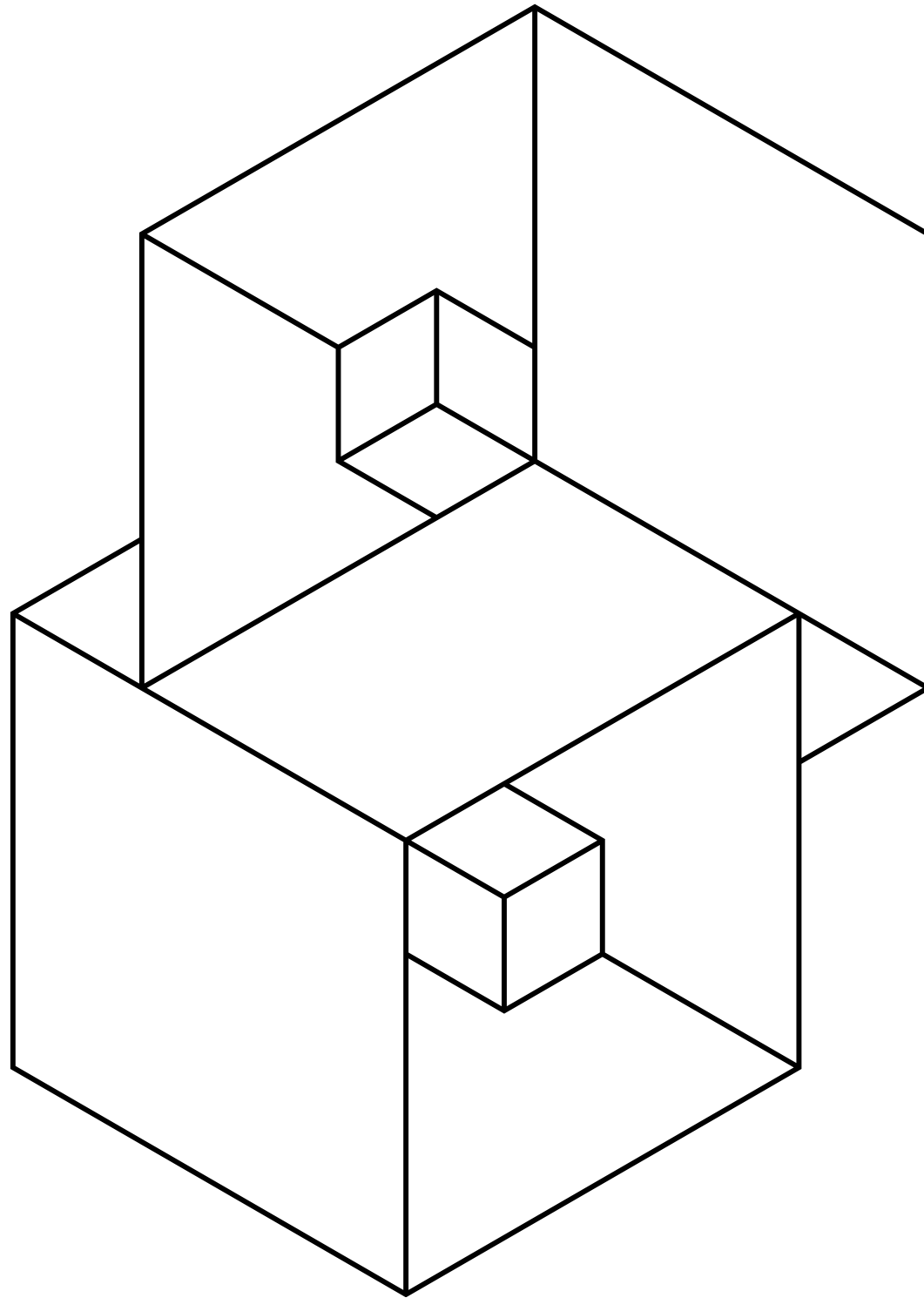




PANDAS



PANDAS



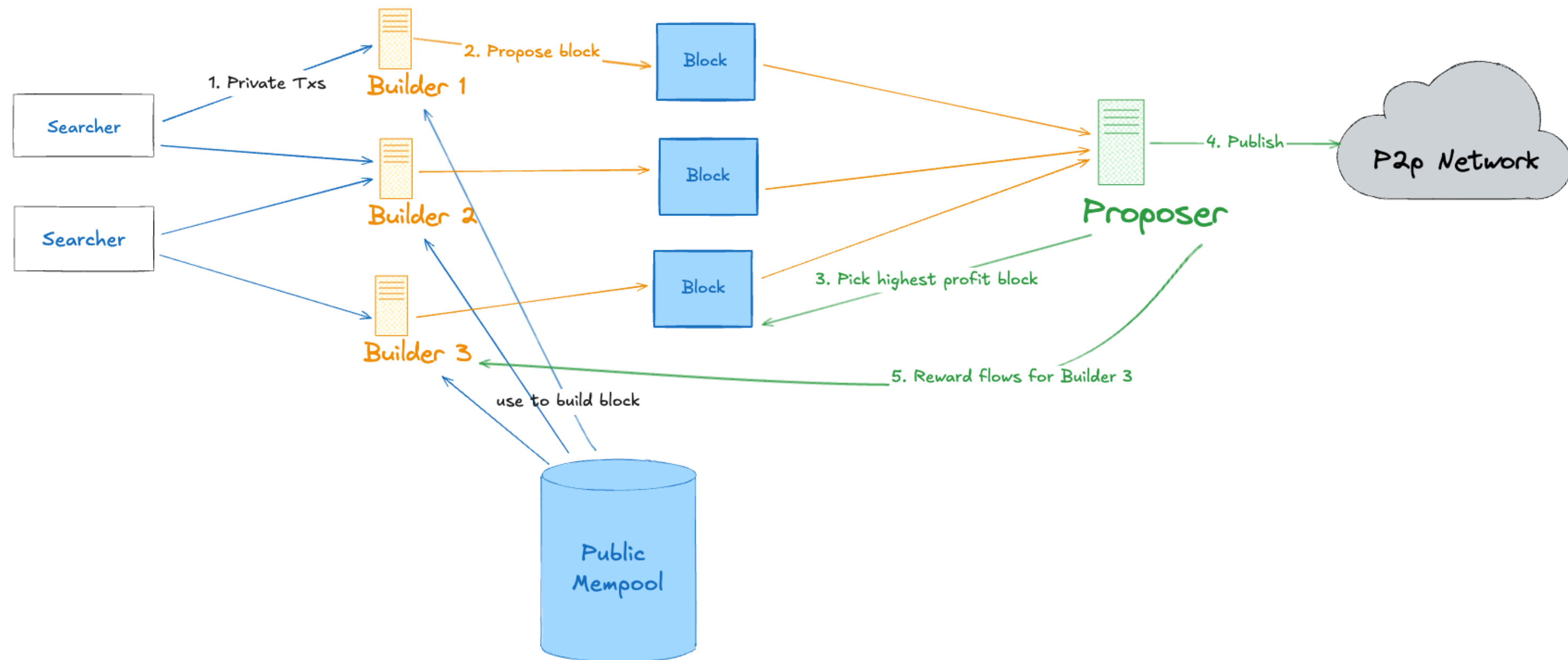
Briefly about how it works ?

PANDAS is a solution that aims to solve the duration of sampling which has to be under 4 seconds

We will discuss about it's architecture and it's cons.

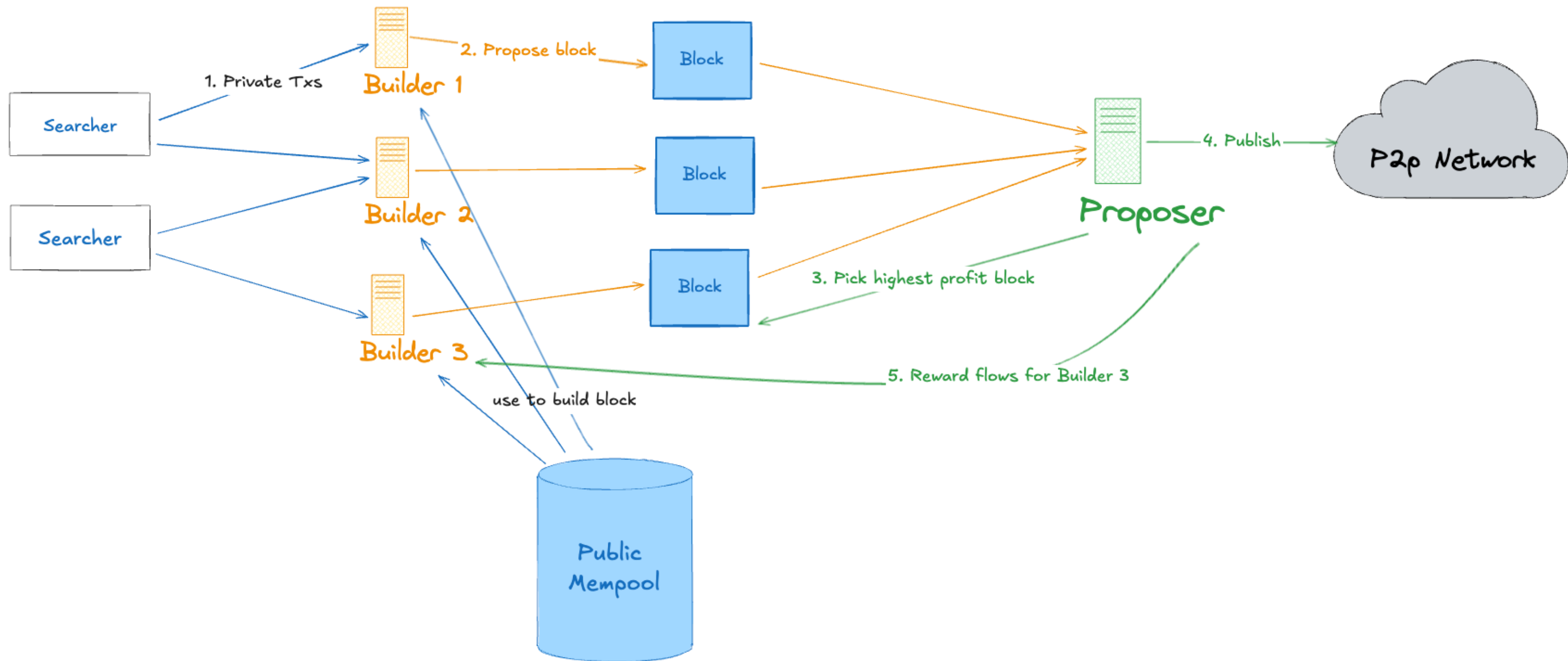
What is Proposer-Builder Separation ?

- +) Building block from transactions, especially in selecting txs to get high profit block required extensive computation => **Require high-bandwidth, high-RAM computer.**
- +) Currently, on Ethereum, 90% validators outsources building block from builders who run MEV-boost client.
- +) When a validator is elected to be proposer, validator will receive blocks from multiple builders. When a block from selected builder is publishing, builder will receive reward.



What is Proposer-Builder Separation ?

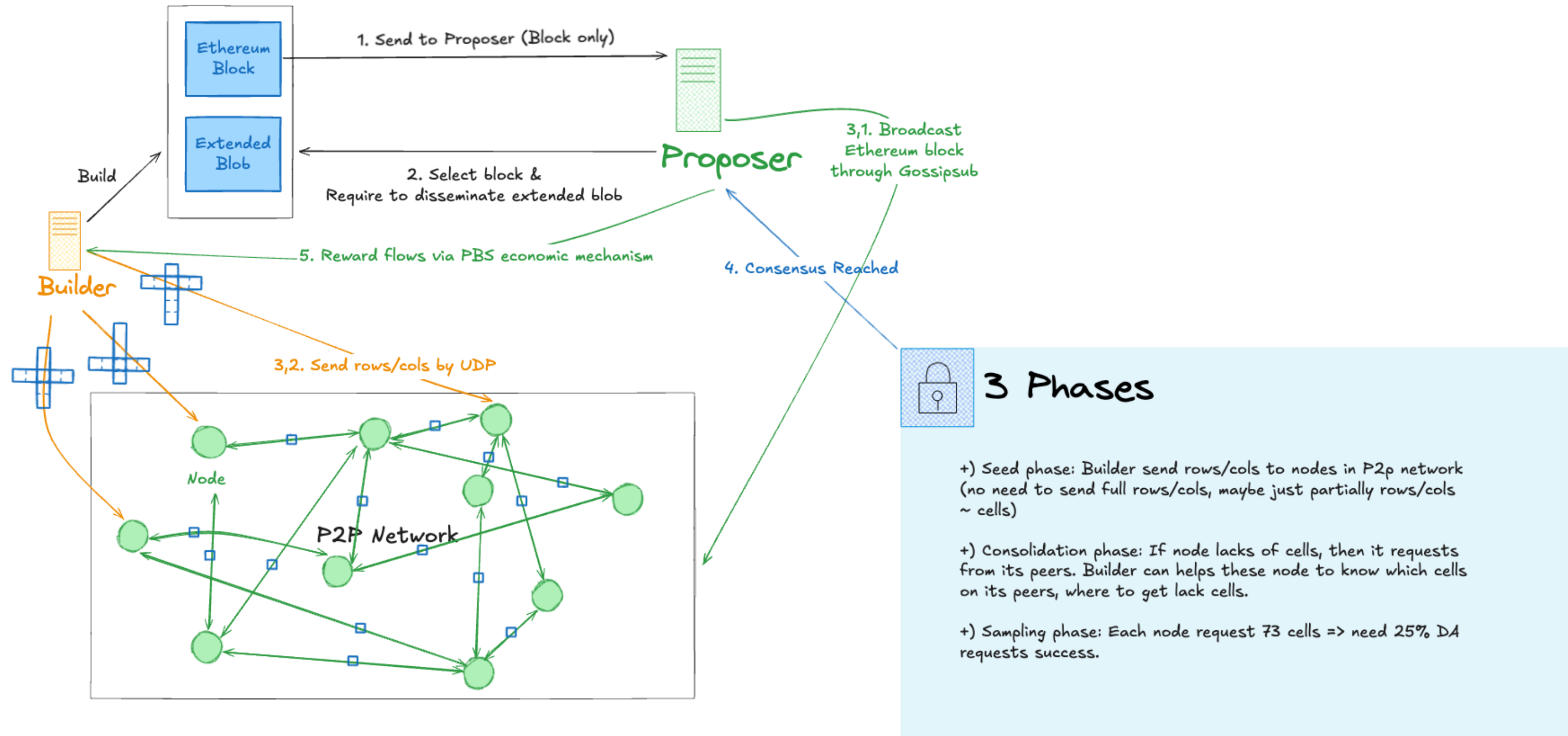
=> PANDAS takes advantages of builders who uses strong computer. Delegates to it more tasks, then validators can keep low-computation computer



Architecture

=> From image, i hope it is easily to understand that builder is delivering assigned rows/cols to nodes

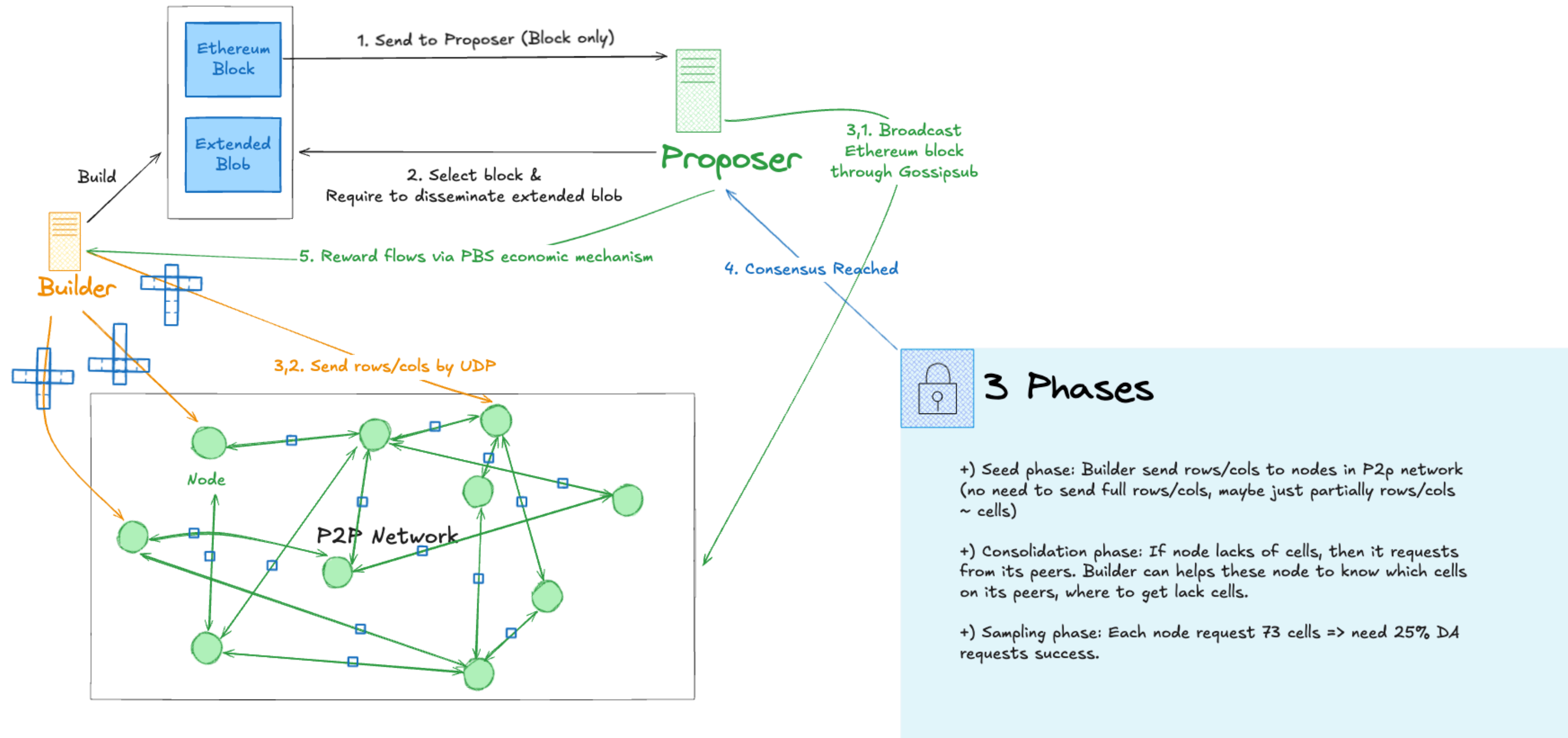
=> Look at three phases, we can realize that downloading is happened before sampling



Problems

As you can see here builder have to know all nodes in the network, so he can deliver rows/cols to nodes in P2P network.

=> This is the disadvantage of PANDAS, also it can only be applied for Ethereum since it has block builder from Flashbot service, same with Solana using Jito.



Overall

1. Advantages:

+) PANDAS has benchmarked to prove that sampling is under 4 seconds, even with block 140MB, size 512 x 512.

2. Disadvantages:

+) Block builder has to know all the node in the network to disseminate rows/cols

+) Block builder is only suitable for large chain like Ethereum, Solana which has this feature. This create a centralized situation in blob propagation

=> Let's look to another one **RDA**